

Abstract

This project presents the complete design, assembly, testing, diagnostics, and validation process of an Electric Two-Wheeler (EV). The objective of the project was to gain practical knowledge of electric vehicle architecture, component integration, electrical systems, and maintenance procedures through hands-on implementation. The assembly process involved the installation of major EV components, including the chassis, BLDC hub motor, battery pack, motor controller, wiring harness, braking system, and auxiliary electrical systems.

After the assembly stage, comprehensive testing and diagnostic procedures were performed to verify electrical continuity, component functionality, system safety, and overall vehicle performance. Troubleshooting techniques and root cause analysis were applied to identify and resolve operational issues during the validation phase. The project provided practical exposure to EV technology, electrical integration, safety practices, technical documentation, and maintenance methodologies. The successful completion of the project enhanced both theoretical understanding and practical skills required for modern electric mobility systems and electric vehicle service operations.